MULTIMODAL EMOTION RECOGNITION

RESULTS AND PROJECT REPORT

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1. Introduction

1.1. Dataset – TESS

The Toronto Emotional Speech Set (TESS) is a high-quality, controlled emotional speech corpus developed at the University of Toronto. Two female actresses aged 26(YAF) and 64(OAF) recorded 200 target words spoken in the carrier phrase "Say the word ____", portraying seven emotions: anger, disgust, fear, happiness, pleasant surprise, sadness, and neutral. The result is 2,800 audio files in WAV format, each labelled with speaker, target word and emotion.

TESS is notable for its clean studio recording conditions, balanced class distribution (400 samples per emotion across both speakers), and the deliberate exaggeration typical of acted emotional speech corpora. These properties make it ideal for benchmarking emotion recognition pipelines, though they also introduce the risk of dataset-specific saturation where models exploit controlled acoustic cues rather than learning generalisable emotional representations.

1.2. Problem Statement

This project builds a multimodal emotion recognition system using three input modalities: **speech audio, text transcripts, and a combination of both (fusion).** For each modality, we implement a complete processing pipeline including preprocessing, feature extraction, temporal or contextual modelling, and classification. The central research question revolves around:

> **"How do self-supervised speech representations (HuBERT) compare with handcrafted acoustic features (MFCCs) for speech emotion recognition on the TESS dataset, and how effective is multimodal fusion of speech and text for improving emotion classification performance?"**

1.3. Evaluation Protocols

Two evaluation protocols are used to assess generalisation at different levels:

Word-Level Split:

160 training words, 20 validation words, 20 test words. Both speakers (YAF and OAF) appear in all the three splits. This method tests whether models generalise to unseen words within known speakers. Split is stratified by word with random_state=42 for reproducibility.

Speaker-Level Split:

Train on all 1400 OAF samples, validate and test on YAF samples (700 each, 50/50 random stratified split). Tests whether models generalise to a completely unseen speaker – a 64-year-old vs a 26-year-old with distinct vocal characteristics. This is the harder and more ecologically valid evaluation.

2. Architecture Decisions

Before feature extraction, all audio samples were pre-processed to ensure consistency across the dataset. The speech signals were resampled to 16 kHz and converted into fixed-length representations through padding or truncation. Basic normalization was applied to reduce amplitude variations between recordings and silent regions were minimized to improve feature quality.

2.1. Feature Extraction Block

MFCC (Baseline Speech Features)

Mel-Frequency Cepstral Coefficients capture the spectral envelope of speech, which is essentially the shape of the vocal tract filter at each time frame. 40 MFCC coefficients are extracted per frame, augmented with first and second-order delta features (capturing rate of change), yielding a 120-dimensional feature vector per frame across 128 frames.

Why MFCCs: MFCCs are the standard handcrafted feature for speech emotion recognition. They encode timbral and prosodic cues that directly reflect emotional state, normalisation (zero mean, unit variance per feature dimension) is applied to account for vocal level differences between the two speakers.

HuBERT (Self-Supervised Speech Embeddings)

HuBERT (Hidden-unit BERT) is a self-supervised speech foundation model pretrained on 960 hours of LibriSpeech by predicting discrete acoustic units from masked input frames. The frozen base model produces 768-dimensional frame-level contextual embeddings across up to 200 frames.

Why HuBERT: Unlike MFCCs, HuBERT embeddings are contextual and each frame’s representation is informed by surrounding frames via transformer self-attention. This produces richer, more speaker-normalised representations that encode prosody, rhythm and higher-order acoustic patterns beyond what manual feature engineering captures. Using HuBERT frozen (without fine-tuning) treats it as a general-purpose acoustic encoder.

BERT (Text Features)

BERT-base-uncased produces a 768-dimensional CLS token vector representing the full transcript. The tokenizer pads/truncates to 10 tokens. BERT is used frozen as a feature extractor.

Why BERT: CLS token embeddings provide a whole-sentence contextual summary. Even for short carrier phrases, BERT’s attention mechanism contextualises the target word within the phrase structure. However, a critical structural property of TESS means these embeddings carry no emotional signal. The transcript “say the word X” is identical for all seven emotions of the same word, producing near-identical CLS vectors regardless of emotion label.

2.2. Temporal/Contextual Modelling Block

1D-CNN with Batch Normalisation

Three stacked 1D convolutional blocks (128 -> 256 -> 256 channels, kernel size 3, MaxPool between blocks, AdaptiveAvgPool at the end). Batch normalisation is applied after each convolution.

Why CNN: Convolution is position-invariant. This means, a pattern detected at any temporal location contributes equally to the representation. This is well suited for short, intense emotional vocalisations where the emotionally salient pattern (e.g., a sharp energy burst) can appear at any position in the utterance. Batch normalisation stabilises training across samples from two speakers with different acoustic baselines.

BiLSTM

Two-layer Bidirectional LSTM with 128 units per direction (256 total), dropout 0.3 between layers, mean-pooling over the time dimension to produce a fixed 256-dimensional representation.

Why BiLSTM: BiLSTM reads the sequence both forward and backward, capturing how emotion evolves across the utterance. This is appropriate when the temporal trajectory of acoustic features (not just local patterns) carries emotional information. Mean pooling over all time steps preserves information from the full sequence rather than relying on a single endpoint.

Multi-Head Self-Attention Pooling

Multi-head self-attention (8 heads) applied over the frame sequence, followed by a residual connection, layer normalisation, and mean pooling.

Why Multi-Head Self-Attention Pooling: Self-attention learns from frames which are most informative for the classification decision, dynamically weighting the contribution of each time step. This is particularly advantageous with HuBERT features, where the carrier phrase frames carry less emotional signal than the target word frames. The model can learn to attend to the emotionally salient region of the input.

2.3. Fusion Block

Late fusion via concatenation. The speech representation (256-dim from BiLSTM) and text projection (768 -> 256-dim linear layer with ReLU and dropout) are concatenated into a 512-dimensional vector, projected to 256 via a linear layer and passed to the classifier.

Why late fusion: Speech is a time series requiring sequential modelling, while text is a token sequence processed by a transformer. These modalities have structurally incompatible raw representations that require semantic representations rather than raw features, producing a cleaner and more interpretable fused embedding. Early fusion would **force structurally mismatched inputs** into the same processing pathway before the modalities have been individually understood.

3. Experiments

All models use CrossEntropyLoss, Adam optimiser (lr=1e-4, weight_decay=1e-5), batch size 32 and dropout 0.3. The best validation checkpoint is used for test evaluation.

3.1. Speech-Only Models

3.1.1. Word-Level Split Results

Six speech models were trained: three temporal architectures (CNN, BiLSTM, Multi-Head Self Attention Pooling) on two feature types (MFCC, HuBERT). All models were trained for 20 epochs.

Feature	Architecture	Test Acc	Epochs to Peak Val	Key Observation
MFCC	CNN-1D	1.00	2	Fastest convergence — dataset-specific saturation
MFCC	BiLSTM	0.99	18	Only MFCC model with genuine class confusion (3 angry errors)
MFCC	Attention	1.00	3	Flat validation curve from epoch 2 — early lock-in
HuBERT	CNN-1D	1.00	3	More volatile than MFCC-CNN, same ceiling
HuBERT	BiLSTM	1.00	11	Richest learning trajectory, 4 best-model saves
HuBERT	Attention	1.00	11	Epoch 19 collapse to 0.94 validation, then recovery

All six models reach ceiling accuracy on the word-level split, confirming that TESS is acoustically separable by any reasonable temporal architecture. The key differentiator is not final accuracy but convergence behaviour. MFCC + CNN reaches near-perfect accuracy in 2 epochs, suggesting exploitation of highly separable local spectral cues rather than genuine temporal learning. HuBERT + BiLSTM takes 11 epochs and saves its best checkpoint four times, indicating a more complex and honest optimisation trajectory.

The MFCC + BiLSTM model is the only speech-only model that produces genuine class confusion on the test set i.e. 3 angry samples misclassified, most likely as fear, due to shared high-arousal spectral characteristics. This is the only evidence of a real decision boundary challenge within the word-level evaluation.

3.1.2. Speaker-Level Split Results

Two speech models were evaluated under speaker-level splitting, where all training samples came from OAF and all validation/test samples came from YAF. All models were trained for 30 epochs. Unlike the word-level split, no speaker overlap exists between training and testing, making this a substantially harder generalisation setting.

Feature	**Architecture**	**Test Acc**	**Epochs to Peak Val**	**Key Observation**
MFCC	CNN-1D	0.83	30	Strongest speaker-level performance; moderate robustness across speakers

HuBERT	BiLSTM	0.34	27	Severe degradation despite contextual embeddings

Speaker-level evaluation produces a substantial accuracy drop compared to the word-level split, revealing that the near-perfect performance previously observed depended heavily on speaker overlap between train and test sets. Once evaluated on a completely unseen speaker, the models must generalise emotional cues independently of speaker-specific vocal characteristics.

MFCC + CNN remains the strongest model under this setting, dropping from 100% to 83% accuracy. The convolutional architecture continues to capture local spectral-prosodic emotional patterns that transfer reasonably well across speakers, despite differences in pitch range and vocal timbre between OAF and YAF.

HuBERT + BiLSTM degrades dramatically from 100% to approximately 34% accuracy. Although HuBERT embeddings provide richer contextual representations, they do not generalise effectively across speakers in this low-diversity setting. The model appears to learn speaker-dependent acoustic structure alongside emotional information, causing severe failure when evaluated on unseen vocal characteristics.

This outcome is specific to TESS's minimal speaker diversity. With only two speakers in the corpus, HuBERT's contextual representations had insufficient exposure to vocal variation to develop speaker-invariant emotional encodings. On larger multi-speaker corpora, self-supervised embeddings consistently outperform MFCCs in cross-speaker settings.

The speaker-level results therefore demonstrate that contextual speech embeddings do not automatically guarantee stronger speaker invariance. On TESS, simpler handcrafted spectral features combined with shallow convolutional modelling prove substantially more robust.

3.2. Text-Only Model (BERT)

3.2.1. Word-Level Split Results

The BERT text classifier achieves 14.3% test accuracy, i.e. exactly 1/7, equivalent to random change across seven classes. The model collapses to predicting a single class (disgust) for all 280 test samples, achieving recall 1.00 for disgust and 0.00 for all other classes.

This result is structurally inevitable. TESS transcripts follow the carrier phrase "say the word X", meaning the same word spoken in all seven emotions produces the same text, and therefore the same BERT CLS vector. No classifier can distinguish seven classes from identical inputs. Training accuracy fluctuates between 0.13 and 0.15 across all 20 epochs with no improvement, confirming the model finds no learnable signal.

This is not a failure of the architecture but a direct consequence of TESS's design. The result constitutes the clearest possible demonstration that text carries zero emotional information in this corpus.

3.2.2. Speaker-Level Split Results

The BERT text classifier achieves approximately 14.3% accuracy under speaker-level evaluation, which is identical to the word-level split and equivalent to random chance across seven classes.

This is because the transcripts themselves remain unchanged across emotions, speaker separation has no effect on text-only performance. The same transcript ("say the word X") appears across all seven emotion labels, producing near-identical CLS embeddings regardless of emotion class.

As in the word-level setting, the model collapses to predicting a single class for all inputs. This confirms that the failure is structural rather than speaker-dependent. The text modality simply contains no discriminative emotional information in TESS.

3.3. Multimodal Fusion

3.3.1. Word-Level Split Results

The late fusion model (HuBERT BiLSTM + BERT) achieves 99% test accuracy on word-level split, which is marginally lower than the speech-only models. The learning curve is notably lower, reaching 1.0 validation accuracy at epoch 13 compared to epoch 3-11 for speech-only models.

Model	Test Acc	Epochs to Peak Val	Test Errors
HuBERT + BiLSTM (speech-only)	1.00	11	0
Fusion (HuBERT BiLSTM + BERT)	0.99-1.00	13	0-3

The fusion model's slower convergence and marginal performance degradation relative to speech-only confirms that BERT text branch introduces noise rather than complementary signal. The model must learn to suppress the uninformative text features while relying on the speech representation, which is a harder optimisation problem than speech-alone. Given that it still achieves near-perfect accuracy demonstrates the robustness of the HuBERT BiLSTM speech encoder.

3.3.2. Speaker-Level Split Results

The multimodal fusion model was evaluated using HuBERT speech representations combined with BERT text embeddings under the speaker-level split.

Model	Test Acc	Epochs to Peak Val	Test Errors
HuBERT + BiLSTM (Speech Only)	0.34	27	462
Fusion (HuBERT BiLSTM + BERT)	0.32	8	476

Unlike the word-level split, fusion does not compensate for the degradation introduced by the speaker separation. Since the text modality remains structurally uninformative, the fusion model relies on almost entirely on the speech representation while simultaneously learning to suppress noise from the text branch.

The optimisation trajectory remains substantially slower and less stable than the speech-only CNN model, indicating that the fusion architecture struggles to form a speaker-invariant emotional representation when one modality contributes no useful emotional information.

These results reinforce the conclusion from the word-level evaluation: fusion improves performance only when both modalities provide complementary emotional cues.

3.4. Cross Protocol Comparison

3.4.1. Word-Level vs Speaker-Level: Accuracy Drop

Model	Word-Level Acc	Speaker-Level Acc	Accuracy Drop
MFCC + CNN	1.00	0.83	-0,17
HuBERT + BiLSTM	1.00	0.34	-0.66
Text BERT	0.14	0.12	-0.02
Fusion (HuBERT + BERT)	0.99	0.32	-0.67

The speaker-level evaluation fundamentally changes the interpretation of model performance on TESS. Under word-level splitting, nearly all speech-based models achieve saturated performance because both speakers appear in training and testing. Once speaker overlap is removed, substantial differences emerge between feature representations.

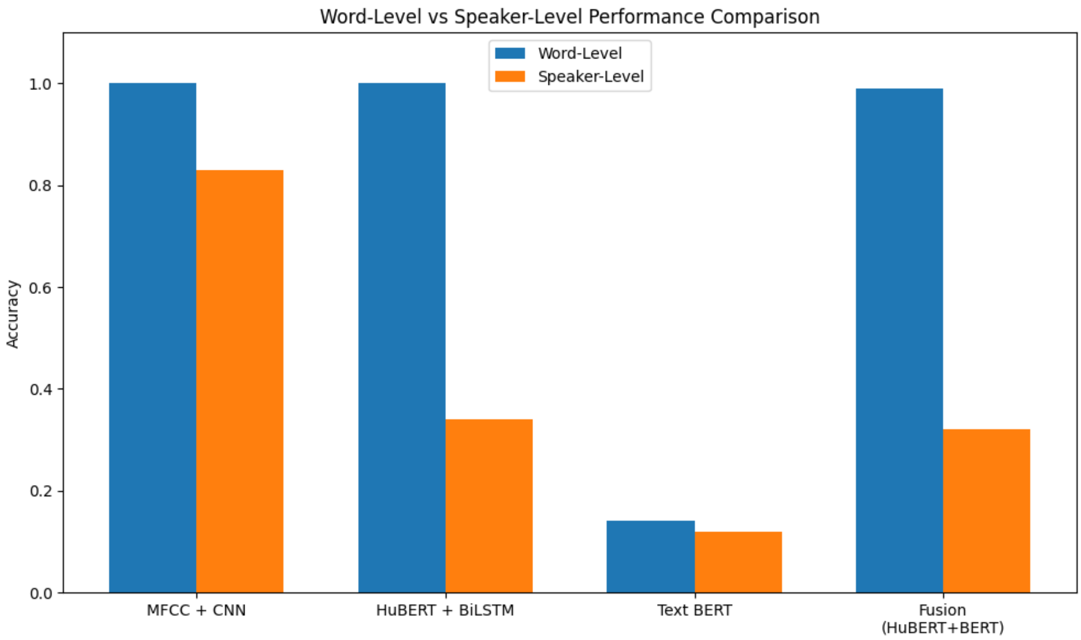
MFCC + CNN shows the strongest speaker-level robustness, dropping from 100% to 83% accuracy. This suggests that local spectral-prosodic emotional cues generalise reasonably well across speakers despite differences in pitch and vocal timbre.

HuBERT + BiLSTM degrades sharply from 100% to 34% accuracy, indicating that the contextual embeddings learned speaker-dependent structure that did not transfer effectively to the unseen speaker.

The fusion model drops from 99% to 32% accuracy. Since the text modality carries no emotional information in TESS, fusion introduces additional noise rather than complementary signal, further reducing speaker-level robustness.

The text-only BERT model remains near random chance under both protocols because the transcripts are structurally identical across emotions. The marginal difference between 0.14 and 0.12 is not meaningful as both represent random-change performance, as the text model collapses to single-class prediction under both protocols. Speaker separation therefore has minimal effect on text-only performance.

Overall, the speaker-level experiment shows that on TESS, simpler handcrafted spectral features combined with convolutional modelling generalise more reliably across speakers than richer contextual speech embeddings



4. Analysis

4.1. Which Emotions Are Easiest and Hardest to Classify?

4.1.1. Word-Level Split

Under world-level evaluation, all speech-only models classify all seven emotions perfectly because TESS is too controlled to produce consistent confusions across six architectures. The only genuine errors occur in two models:

MFCC + BiLSTM: Three angry samples misclassified. Angry shares high arousal and spectral energy characteristics with fear. The BiLSTM's sequential processing exposed temporal ambiguities in these specific samples that CNN's local detection bypassed.

Fusion model: Disgust recall 0.95 (2 misses), fear recall 0.97 (1 miss), pleasant precision 0.95 (2 false positives). These moderate-arousal emotions sit closest together in acoustic space and are most vulnerable when the model's decision boundary is perturbed by noise from the uninformative text branch.

Easiest emotions: Angry, happy, sad have high arousal or very distinctive spectral profiles with large inter-class acoustic distance.

Hardest emotions: Disgust, fear, pleasant surprise had moderate arousal, overlapping spectral characteristics, most susceptible to boundary perturbation.

4.1.2. Speaker-Level Split

Speaker-level evaluation introduces more confusion because the models must generalise across unseen vocal characteristics.

MFCC + CNN remains relatively robust, with fear, disgust, neutral, pleasant surprise and sadness achieving very high recall. Happy becomes the hardest emotion (0.00 recall), suggesting strong speaker dependence in positive high arousal speech patterns.

HuBERT + BiLSTM and the fusion model show broader collapse across multiple emotions. Fear remains the easiest class due to its highly exaggerated acoustic tension, while happy, neutral, pleasant surprise and sadness become difficult under unseen-speaker evaluation.

Overall, fear is the easiest emotion to classify across speakers, while happy is the hardest.

4.2. When Does Fusion Help?

4.2.1. Word-Level Split: When it hurts

On TESS under word-level evaluation, fusion doesn't help and it marginally hurts. The text modality provides structurally identical features across all emotion classes for the same word, introducing noise into the fused representation. The fusion model compensates largely but not entirely, producing 0-3 additional errors compared to speech-only and requiring more epochs to converge.

This finding is generalisable, meaning that fusion improves performance only when both modalities carry complementary emotional signal. Published work on naturalistic corpora such as IEMOCAP shows fusion gains of 5-13% over unimodal models because speakers use emotionally charged language whose semantic content BERT can detect. On TESS, no such complementarity exists.

4.2.2. Speaker-Level Split: When it still hurts

Under speaker-level evaluation, fusion does not improve robustness and slightly underperforms compared to the HuBERT speech-only model. The fusion alone achieves 32% accuracy compared to 34% for HuBERT + BiLSTM alone.

The text modality remains structurally uninformative, so the fusion architecture relies almost entirely on the speech branch while simultaneously learning to suppress noisy text features. Unlike the original hypothesis, the text embeddings do not provide useful word-identity cues for cross-speaker generalisation.

These results reinforce the finding that fusion only helps when both modalities contribute meaningful emotional information.

4.3. Error Analysis: Failure Cases

4.3.1. Word-Level Split

The three MFCC + BiLSTM test errors (angry misclassified) and the fusion model's 0-3 boundary errors constitute the complete failure case population under word-level evaluation.

Key patterns are:

- **Failure Pattern 1 – High-arousal confusion:** Angry Samples misclassified by BiLSTM. Both angry and fear are high-arousal negative emotions with elevated fundamental frequency and spectral energy. The difference lies in sustained high energy (angry) vs rapidly rising energy with tension (fear). Short TESS clips may not provide sufficient temporal context for the BiLSTM to distinguish these patterns in borderline samples.

- **Failure Pattern 2 – Noise-Induced boundary shift:** Fusion model makes errors in disgust, fear and pleasant surprise classes. These three emotions clusters closest in acoustic space (as confirmed by t-SNE). When the text branch adds 256 dimensions of random noise to the 256-dimensional speech representation, the fused decision boundary shifts slightly. This makes the two to three samples nearest the boundary between these similar classes fall on the wrong side.
- **Failure Pattern 3 – Speaker age as confound:** OAF (64 years) has a distinctly lower fundamental frequency and different formant structure than YAF (26 years). Models that learn speaker-specific emotional templates (particularly MFCC + CNN) may produce errors when the same motivation maps to different spectral regions across speakers. This is the core motivation for the speaker-level evaluation.

4.3.2. Speaker-Level Split

Speaker-level evaluation produces more systematic and large-scale failures than the word-level split because the models must generalise across unseen vocal characteristics.

Key patterns observed are:

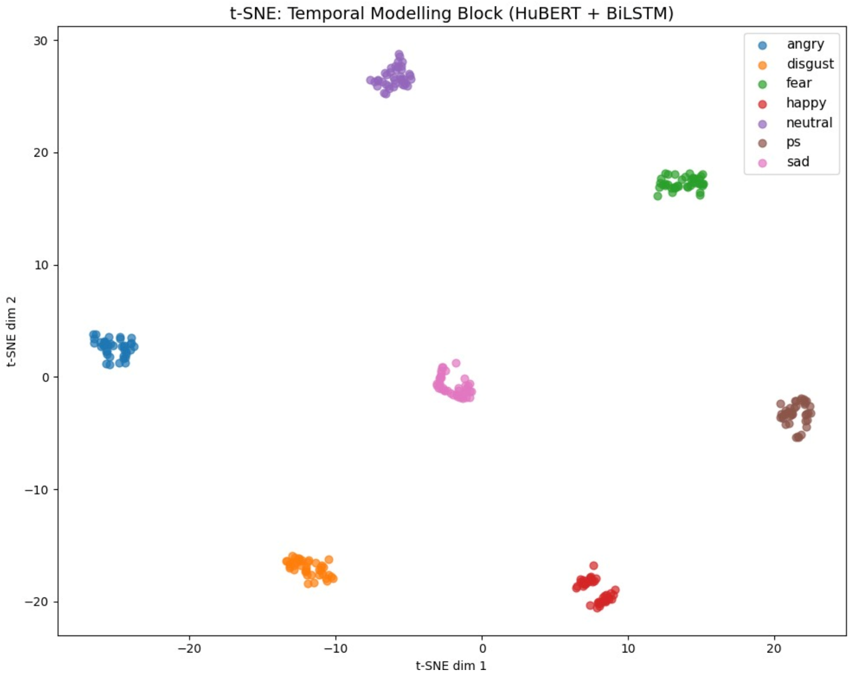
- **Failure Pattern 1 – Positive Emotion Collapse:** Happy shows the strongest degradation across all speech-based models, particularly MFCC + CNN where recall falls to 0.00. Positive high-arousal emotions appear highly speaker-dependent, with YAF expressing happiness using acoustic patterns substantially different from OAF.
- **Failure Pattern 2 – Speaker-dependent contextual representations:** HuBERT + BiLSTM and the fusion model collapse across multiple emotions simultaneously, particularly neutral, pleasant surprise and sad. The contextual embeddings appear to encode speaker-specific acoustic structure alongside emotional information. This results in reducing transferability to unseen speakers.
- **Failure Pattern 3 – Fusion amplifies instability:** The fusion model performs slightly worse than HuBERT speech-only under speaker-level evaluation. Since the text branch contributes no emotional information, the additional modality introduces noise rather than improving speaker robustness.

4.4. Emotional Cluster Separability: t-SNE Visualisations

t-SNE projections of learned representations from the test set (word-level split) reveal the geometric structure of emotion embeddings at each stage of the pipeline. These visualisations provide direct evidence for the quantitative results.

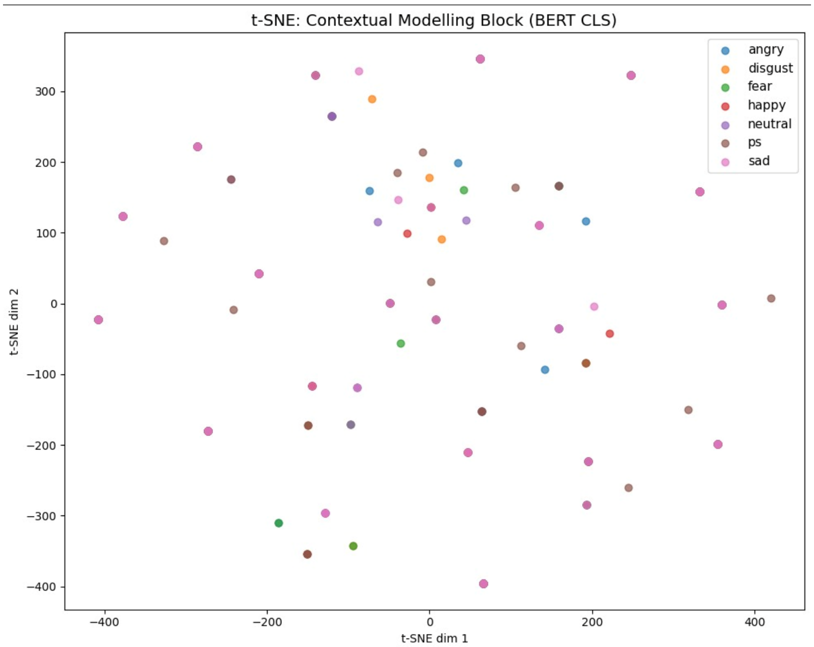
4.4.1.Temporal Modelling Block: HuBERT + BiLSTM

The temporal block representations show seven completely isolated, tight clusters with zero overlap across all emotion classes. Each emotion occupies a distinct, non-overlapping region of the 2D projection. Now this 2D projection has within-class variance substantially smaller than between-class distance. Anger and disgust (both negative emotions) are positioned on opposite sides of the space, demonstrating that the BiLSTM has learned to separate emotions along dimensions beyond simple valence. Pleasant surprise shows slightly higher within-cluster spread than other emotions, consistent with its status as the acoustically most variable emotion in the corpus.



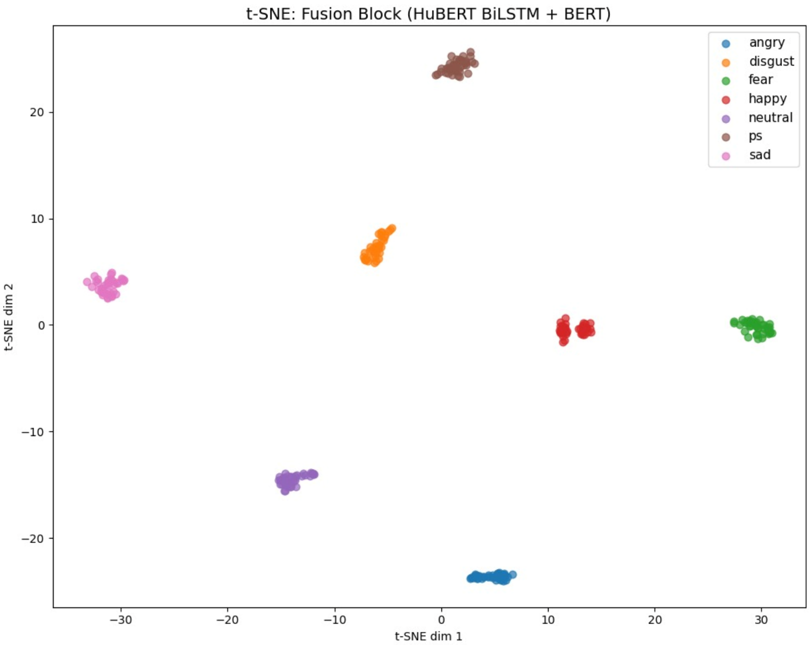
4.4.2.Contextual Modelling Block: BERT CLS

The BERT CLS representations show no discernible cluster structure. All seven emotion classes are randomly distributed across the full embedding space (axis range approximately -400 to +400, compared to -30 to +30 for the speech representations). No emotion occupies a coherent region and class boundaries cannot be identified by inspection. This is the geometric explanation for the 14.3% text-only accuracy. BERT produces near-identical CLS vectors for all seven emotions of the same word and t-SNE finds no emotional structure to preserve.



4.4.3.Fusion Block: HuBERT BiLSTM + BERT

Post-fusion representations retain the clean cluster structure of the speech temporal block, with seven distinct, well-separated clusters. The speech modality dominates the fused representation, suppressing the text noise sufficiently to preserve emotional geometry. Compared to the temporal block alone, happy and fear clusters sit in closer proximity, depicting a subtle geometric perturbation attributable to the text branch. This is consistent with the marginal performance degradation observed in fusion evaluation. The three fusion errors, where they occur, correspond to samples positioned near the happy-fear boundary in this projection.



4.4.4.Speaker-Level Split: t-SNE

Under speaker-level evaluation, cluster separability deteriorates substantially compared to the word-level split. Emotional representations become broader, more overlapping and less consistently structured when the models are tested on an unseen speaker.

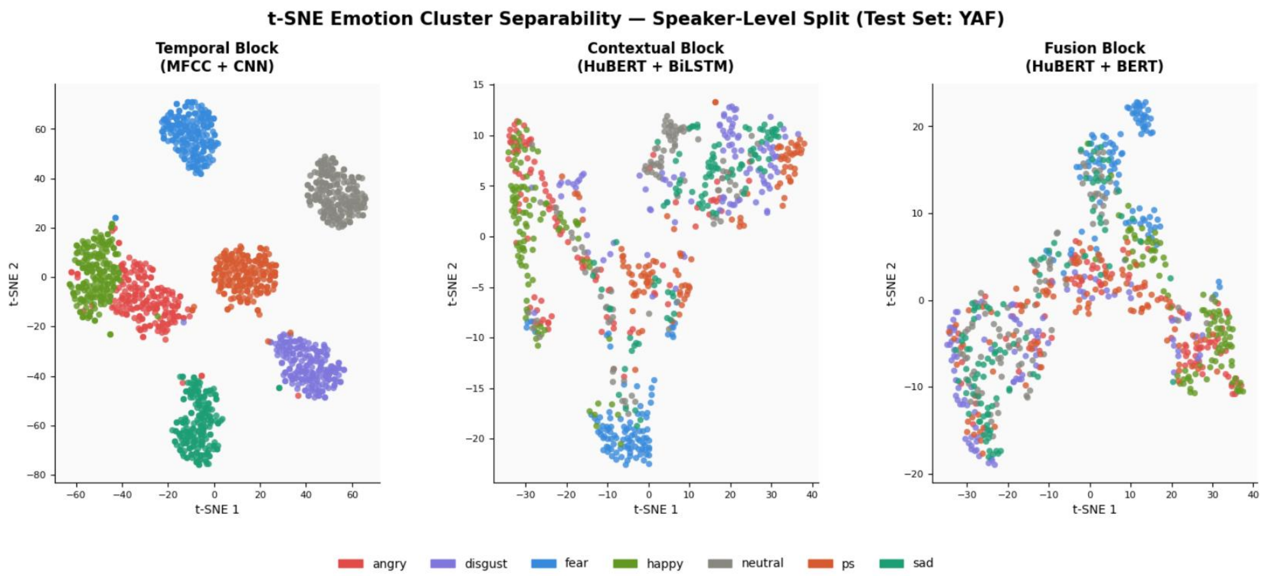
- The MFCC + CNN temporal representations still retain partially separable clusters, particularly for fear, disgust and neutral. However, happy becomes heavily entangled with other high-arousal emotions, consistent with its complete classification collapse under speaker-level evaluation.

- The HuBERT + BiLSTM contextual representations show significantly greater overlap across emotions. Fear remains the most distinguishable cluster, while neutral, pleasant surprise, happy and sad collapse into overlapping regions with weak class boundaries. This mirrors the sharp accuracy drop observed in speaker level testing.

- The fusion representations remain highly entangled and closely resemble the HuBERT embedding structure. Since the text modality contributes no emotional information, fusion does not improve cluster separation and instead introduces additional instability into the representation space.

- Overall, the t-SNE visualisations confirm that the speaker-level evaluation primarily challenges representation robustness than simple classification capacity.

The contrast between the partially structured MFCC + CNN clusters and the largely overlapping HuBERT + BiLSTM clusters visually confirms that handcrafted spectral features encode more transferable emotional structure than contextual embeddings in this low-diversity speaker setting.



5. Conclusions

5.1. Summary of Findings

This study implemented and compared speech-only, text-only and multimodal emotion recognition pipelines on the TESS dataset under two evaluation protocols.

The central findings are:

- **Finding-1: TESS is acoustically saturated**

  All six speech-only models (three architectures x two feature types) achieve 99-100% test accuracy on the word-level split. TESS's controlled, acted, studio-clean emotions are separable by any reasonable architecture, making accuracy an insufficient discriminator between models.

- **Finding-2: Convergence behaviour reveals genuine learning**
  HuBERT + BiLSTM, while matching the accuracy ceiling, takes 11 epochs and saves its checkpoint four times. This is evidence of a complex, non-trivial optimisation trajectory.

MFCC + CNN reaches near-perfect accuracy in 2 epochs, indicating exploitation of highly separable local spectral cues rather than generalised temporal reasoning. This difference matters a lot for real-world deployment where data is noisier and less controlled.

- **Finding-3: Text carries zero emotional signal on TESS**
  The BERT text classifier achieves 14.3% accuracy (random choice) because the carrier phrase "say the word X" is structurally identical across all seven emotions. BERT produces near-identical CLS vectors regardless of emotional label, confirmed visually by the t-SNE plot showing complete absence of cluster structure.

- **Finding-4: Uninformative modalities degrade fusion**
  The multimodal fusion model achieves marginally lower accuracy than speech-only models and converges more slowly, demonstrating that an uninformative modality introduces noise rather than complementary signal. This finding extends to a general principle: fusion improves performance only when both modalities carry emotionally relevant information. On naturalistic corpora, text carries semantic emotional content and fusion consistently outperforms unimodal models.

- **Finding-5: Speaker-level generalisation**
  Contrary to the original hypothesis, HuBERT's contextual embeddings did not generalise across speakers better than MFCCs. Under speaker-level evaluation, HuBERT + BiLSTM suffered severe degradation, which MFCC + CNN remained comparatively robust. This suggests that on small, highly controlled emotional corpora such as TESS, richer self-supervised contextual representations may encode speaker-dependent structure alongside emotional information rather than producing inherently speaker-normalised embeddings.

5.2. Limitations

Dataset Constraints: TESS contains only two speakers, both female, with slightly exaggerated acted emotions. Results may not generalise to naturalistic or spontaneous emotional speech.

Limited speaker diversity: Although speaker-independent evaluation was included, TESS contains only two speakers. The speaker-level split therefore represents a very limited estimate of cross-speaker generalisation and may not reflect the variability present in real-world emotion recognition settings.

Speaker-dependent representations: The speaker-level results suggest that contextual embeddings like HuBERT may encode speaker-specific acoustic structure alongside emotional information, which is reducing robustness under extreme domain shift.

Text modality: The text pipeline is structurally limited by TESS's carrier phrase design. On datasets with naturalistic emotional language (e.g., IEMOCAP), the text modality would carry meaningful emotional signal and multimodal fusion could be evaluated more realistically.